EXPERIMENT 08

Creating Custom Workflows, Validations, and Automated Triggers in Zoho Creator

Library Book Issue Request Form (Workflow Automation Use Case)

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1. Aim

To design and implement custom form workflows, field validations, schedule triggers, and automated notification actions within a Zoho Creator application (Library Book Issue Request Form) to streamline book borrowing and approval processes.

The experiment covers form-level validation rules, on-user-input triggers, on-success workflows, scheduled actions, and automated email/SMS dispatchers.

2. Procedure

Step 1: Define workflow triggers and events. Identified key execution points in the library request lifecycle: field entry, form validation prior to submission, successful record creation, and scheduled daily due-date checks.

Step 2: Configure field validation rules. Added Deluge script validations to ensure requested return dates are after issue dates and that student roll numbers match active enrollment records.

Step 3: Build on-success workflows. Set up automated record updates to decrement available book inventory counts upon successful request submission.

Step 4: Implement approval workflows. Configured multi-step approval routing sending task notifications to library staff whenever a new book request is submitted.

Step 5: Schedule periodic automated tasks. Created scheduled script blocks to check for overdue books daily and automatically trigger fine calculation reminders.

Note: This is a structured workflow automation exercise. Script logic and trigger rules follow standard Zoho Creator Deluge execution standards.

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3. Output

3.1 Workflow Trigger Matrix & Automation Logic

The automation matrix outlines trigger conditions and corresponding actions across the library workflow:

Trigger Type	Execution Point	Condition / Rule	Automated Action
Field Validation	On User Input	Return Date > Issue Date	Display validation error if date is invalid.
Form Submission	On Success	Book Request Saved	Decrement book stock and notify librarian.
Approval Routing	Status Change	Approval Status = Approved	Send confirmation email to student.
Scheduled Task	Daily at Midnight	Current Date > Due Date	Calculate overdue fine and dispatch reminder.

3.2 Deluge Validation Script Implementation (Illustrative)

```
// On Validate workflow for Book Issue Form
if (input.Return_Date <= input.Issue_Date)
{
  alert "Return date must be strictly after the issue date.";
  cancel submit;
}
```

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3.3 Automated Notification & Exception Management

Automated email handlers were configured to dispatch HTML-formatted confirmation slips containing book details, issue dates, and barcode identifiers immediately upon approval. Exception handlers catch invalid inventory states to prevent negative stock counts.

4. Final Result

Successfully designed and implemented custom workflows, field validations, approval routing, and scheduled automated triggers for the Library Book Issue Request Form application in Zoho Creator.

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